Past Problems

for the International Mathematical Olympiad

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Part I. Competition Problems

Problem 2010.1. Determine all functions $f\colon \mathbb{R} \to \mathbb{R}$ for which

$$f(\lfloor x \rfloor y) = f(x)\lfloor f(y) \rfloor$$

for every $x, y \in \mathbb{R}$.

Problem 2010.2. Let I be the incentre of triangle ABC, and let Γ be the circumcircle of ABC. The line AI meets Γ again at D. Choose E on the arc $\widehat{BDC}$ and F on the side BC such that

$$\angle BAF = \angle CAE < \frac{1}{2}\angle BAC.$$

Let G be the midpoint of IF. Prove that the lines DG and EI intersect at a point of Γ.

Problem 2010.3. Find all functions $g\colon \mathbb{N} \to \mathbb{N}$ such that

$$(g(m) + n)(m + g(n))$$

is a perfect square for every $m, n \in \mathbb{N}$.

Problem 2010.4. Let P lie inside triangle ABC. The lines AP, BP, and CP meet the circumcircle Γ of ABC again at K, L, and M, respectively. The tangent to Γ at C meets AB at S. If $SC = SP$, prove that $MK = ML$.

Problem 2010.5. Initially, each of six boxes $B_1, B_2, \ldots, B_6$ contains one coin. The following operations are permitted:

(1) Choose a nonempty box B_j, where $1 \leq j \leq 5$. Remove one coin from B_j and place two coins in B_{j+1}.

(2) Choose a nonempty box B_k, where $1 \leq k \leq 4$. Remove one coin from B_k, then exchange the contents of B_{k+1} and B_{k+2}, which may be empty.

Determine whether a finite sequence of these operations can leave B_1, B_2, B_3, B_4, B_5 empty and leave exactly $2010^{(2010^{2010})}$ coins in B_6.

Problem 2010.6. Let $(a_n)_{n\geq 1}$ be a sequence of positive real numbers. Suppose that there is a positive integer s such that

$$a_n = \max\{a_k + a_{n-k} \mid 1 \leq k \leq n-1\}$$

for every $n > s$. Prove that there exist positive integers ℓ, N with $\ell \leq s$ such that

$$a_n = a_\ell + a_{n-\ell}$$

for all $n \geq N$.

Problem 2011.1. For a set $A = \{a_1, a_2, a_3, a_4\}$ of four distinct positive integers, put $s_A = a_1 + a_2 + a_3 + a_4$, and let n_A be the number of pairs (i, j) with $1 \leq i < j \leq 4$ for which $a_i + a_j$ divides s_A. Determine all such sets A for which n_A is as large as possible.

Problem 2011.2. Let S be a finite set of at least two points in the plane, with no three collinear. A *windmill* begins with a line ℓ passing through exactly one point $P \in S$. Rotate ℓ clockwise about P until it first passes through another point $Q \in S$. The point Q becomes the pivot, and the line continues rotating clockwise about Q until it reaches another point of S. Repeat this process indefinitely.

Prove that P and ℓ can be chosen so that every point of S becomes a pivot infinitely many times.

Problem 2011.3. Suppose that $f\colon \mathbb{R} \to \mathbb{R}$ satisfies

$$f(x + y) \leq yf(x) + f(f(x))$$

for all $x, y \in \mathbb{R}$. Prove that $f(x) = 0$ whenever $x \leq 0$.

Problem 2011.4. A balance and n weights of masses $2^0, 2^1, \ldots, 2^{n-1}$ are given, where n is a positive integer. Place the weights on the balance one at a time, choosing at each step an unused weight and putting it on either pan. The right pan must never be heavier than the left pan, including after each placement.

How many ways are there to place all n weights under this restriction?

Problem 2011.5. Let $f\colon \mathbb{Z} \to \mathbb{N}$ be a function such that $f(m - n)$ divides $f(m) - f(n)$ for every $m, n \in \mathbb{Z}$. Prove that whenever $m, n \in \mathbb{Z}$ satisfy $f(m) \leq f(n)$, we have $f(m) \mid f(n)$.

Problem 2011.6. Let ABC be an acute triangle with circumcircle Γ. Take a tangent line ℓ to Γ, and reflect ℓ in the lines BC, CA, and AB to obtain ℓ_a, ℓ_b, and ℓ_c, respectively. Prove that the circumcircle of the triangle formed by ℓ_a, ℓ_b, and ℓ_c is tangent to Γ.

Problem 2012.1. In triangle ABC, let J be the centre of the excircle opposite A. This excircle touches BC at M, and the lines AB and AC at K and L, respectively. Let the lines LM and BJ intersect at F, and the lines KM and CJ at G. Let the lines AF and BC intersect at S, and the lines AG and BC at T. Prove that M is the midpoint of ST.

The excircle opposite A is tangent to the segment BC, the ray AB beyond B, and the ray AC beyond C.

Problem 2012.2. Let $n \geq 3$ be an integer and let $a_2, a_3, \ldots, a_n$ be positive real numbers with $a_2 a_3 \cdots a_n = 1$. Prove that

$$(1 + a_2)^2 (1 + a_3)^3 \cdots (1 + a_n)^n > n^n.$$

Problem 2012.3. In the *liar's guessing game*, two players A and B know positive integers k and n. Player A secretly chooses integers x, N with $1 \le x \le N$ and tells B the true value of N, but not x. Player B may repeatedly name any set S of positive integers and ask whether $x \in S$, including repeating a previous question. Player A answers each question immediately with yes or no; lying is allowed, provided that among every $k+1$ consecutive answers at least one is truthful.

After any number of questions, B chooses a set X of at most n positive integers. Player B wins exactly when $x \in X$. Prove that:

(a) If $n \ge 2^k$, then B can guarantee a win.

(b) For all sufficiently large k, some integer $n \ge 1.99^k$ does not allow B to guarantee a win.

Problem 2012.4. Determine all functions $f\colon \mathbb{Z} \to \mathbb{Z}$ satisfying

$$f(a)^2 + f(b)^2 + f(c)^2 = 2f(a)f(b) + 2f(b)f(c) + 2f(c)f(a)$$

for every triple of integers a, b, c with $a + b + c = 0$.

Problem 2012.5. Let ABC be a right triangle with $\angle BCA = 90^\circ$, and let D be the foot of the altitude from C. Choose X in the interior of CD. Let K be on the segment AX with $BK = BC$, and L on the segment BX with $AL = AC$. If the lines AL and BK meet at M, prove that $MK = ML$.

Problem 2012.6. Determine all positive integers n for which some nonnegative integers $a_1, \ldots, a_n$ satisfy

$$\frac{1}{2^{a_1}} + \frac{1}{2^{a_2}} + \cdots + \frac{1}{2^{a_n}} = \frac{1}{3^{a_1}} + \frac{2}{3^{a_2}} + \cdots + \frac{n}{3^{a_n}} = 1.$$

Problem 2013.1. Given positive integers k, n, prove that some positive integers $m_1, \ldots, m_k$, not necessarily distinct, satisfy

$$1 + \frac{2^k - 1}{n} = \prod_{i=1}^{k} \left(1 + \frac{1}{m_i}\right).$$

Problem 2013.2. A configuration of 4027 points in the plane is called *Colombian* if 2013 of its points are red and 2014 are blue, with no three collinear. A collection of lines is *good* for such a configuration if

(1) none of its lines passes through a point of the configuration;

(2) every region into which the lines divide the plane contains points of at most one colour.

Find the smallest k such that every Colombian configuration admits a good collection of k lines.

Problem 2013.3. Let the excircle of triangle ABC opposite A touch BC at A_1. Let B_1 and C_1 be the analogous tangency points on the sides CA and AB for the excircles opposite B and C. Suppose that the circumcentre of $A_1B_1C_1$ lies on the circumcircle of ABC. Prove that ABC is a right triangle.

Here the excircle opposite A touches the segment BC and the rays AB, AC beyond B, C, respectively; the other excircles are defined analogously.

Problem 2013.4. Let ABC be an acute triangle with orthocentre H, and let W lie strictly between B and C on the side BC. Let M and N be the feet of the altitudes from B and C, respectively. Let ω_1 be the circumcircle of triangle BWN, and let $X \in \omega_1$ be such that WX is a diameter. Likewise, let ω_2 be the circumcircle of CWM, and let $Y \in \omega_2$ be such that WY is a diameter. Prove that X, Y, H are collinear.

Problem 2013.5. Suppose that $f\colon \mathbb{Q}_{>0} \to \mathbb{R}$ satisfies

(i) $f(x)f(y) \geq f(xy)$ for all $x, y \in \mathbb{Q}_{>0}$;
(ii) $f(x+y) \geq f(x) + f(y)$ for all $x, y \in \mathbb{Q}_{>0}$;
(iii) $f(a) = a$ for some rational number $a > 1$.

Prove that $f(x) = x$ for every $x \in \mathbb{Q}_{>0}$.

Problem 2013.6. Mark $n+1$ equally spaced points on a circle, where $n \geq 3$. Label them with $0, 1, \ldots, n$, each used once, and consider two labellings equivalent if one is obtained from the other by rotating the circle. A labelling is *beautiful* if, whenever $a < b < c < d$ and $a + d = b + c$, the chord joining labels a and d does not intersect the chord joining labels b and c.

Let M count the beautiful labellings up to rotation, and let N count ordered pairs (x, y) of positive integers with $x + y \leq n$ and $\gcd(x, y) = 1$. Prove that $M = N + 1$.

Problem 2014.1. Let $a_0 < a_1 < a_2 < \cdots$ be an infinite sequence of positive integers. Prove that there is exactly one integer $n \geq 1$ for which

$$a_n < \frac{a_0 + a_1 + \cdots + a_n}{n} \leq a_{n+1}.$$

Problem 2014.2. Let $n \geq 2$ be an integer. A placement of n rooks on an $n \times n$ chessboard is called *peaceful* if each row and each column contains exactly one rook. Determine the largest positive integer k such that every peaceful placement contains a $k \times k$ square of unit cells with no rook in any of its cells.

Problem 2014.3. Let $ABCD$ be a convex quadrilateral with $\angle ABC = \angle CDA = 90^\circ$. Let H be the foot of the perpendicular from A to BD. Points S and T lie on the sides AB and AD, respectively, with H inside triangle SCT and

$$\angle CHS - \angle CSB = 90^\circ, \qquad \angle THC - \angle DTC = 90^\circ.$$

Prove that the line BD is tangent to the circumcircle of triangle TSH.

Problem 2014.4. Let P and Q lie on the side BC of an acute triangle ABC, with $\angle PAB = \angle BCA$ and $\angle CAQ = \angle ABC$. Choose M on the line AP and N on the line AQ so that P and Q are the midpoints of AM and AN, respectively. Prove that the lines BM and CN intersect on the circumcircle of ABC.

Problem 2014.5. For every positive integer n, the Bank of Cape Town issues coins worth $1/n$. Consider any finite collection of these coins, with denominations allowed to repeat, whose total value is at most $99 + \frac{1}{2}$. Prove that the coins can be divided into at most 100 groups, each having total value at most 1.

Problem 2014.6. A set of lines in the plane is in *general position* if no two are parallel and no three pass through one point. The lines divide the plane into regions; those with finite area are called *finite regions*. Prove that, for all sufficiently large n, any n lines in general position have at least $\sqrt{n}$ lines that can be coloured blue so that no finite region has a boundary made entirely of blue lines.

Problem 2015.1. A finite set S of points in the plane is called *balanced* if, for every pair of distinct points $A, B \in S$, some $C \in S$ satisfies $AC = BC$. It is called *centre-free* if no point $P \in S$ satisfies $PA = PB = PC$ for three distinct points $A, B, C \in S$.

(a) Prove that a balanced set of n points exists for every integer $n \geq 3$.

(b) Determine all integers $n \geq 3$ for which a balanced centre-free set of n points exists.

Problem 2015.2. Find all triples (a, b, c) of positive integers such that $ab - c$, $bc - a$, and $ca - b$ are all powers of 2.

Problem 2015.3. Let ABC be an acute triangle with $AB > AC$, circumcircle Γ, orthocentre H, and altitude foot F from A. Let M be the midpoint of BC. Let Q be the point on Γ satisfying $\angle HQA = 90^\circ$, and let K be the point on Γ satisfying $\angle HKQ = 90^\circ$. Assume that A, B, C, K, Q are distinct and appear on Γ in that order.

Prove that the circumcircles of KQH and FKM are tangent.

Problem 2015.4. Let Ω be the circumcircle of triangle ABC, with circumcentre O. A circle Γ centred at A meets segment BC at distinct points D, E in the order B, D, E, C. Let F, G be the intersections of Γ with Ω such that A, F, B, C, G occur in this order on Ω. The circumcircle of BDF meets segment AB again at K, and the circumcircle of CGE meets segment CA again at L.

Suppose that lines FK and GL are distinct and intersect at X. Prove that X lies on AO.

Problem 2015.5. Find all functions $f\colon \mathbb{R} \to \mathbb{R}$ for which

$$f(x + f(x + y)) + f(xy) = x + f(x + y) + yf(x)$$

for every $x, y \in \mathbb{R}$.

Problem 2015.6. An integer sequence $(a_j)_{j \geq 1}$ satisfies

(i) $1 \leq a_j \leq 2015$ for every $j \geq 1$;

(ii) $k + a_k \neq \ell + a_\ell$ whenever $1 \leq k < \ell$.

Prove that there exist positive integers b, N such that

$$\left| \sum_{j=m+1}^{n} (a_j - b) \right| \leq 1007^2$$

for all integers $n > m \geq N$.

Problem 2016.1. Triangle BCF is right-angled at B. Choose A on line CF so that $FA = FB$ and F lies between A and C. Point D satisfies $DA = DC$, with AC the bisector of $\angle DAB$. Point E satisfies $EA = ED$, with AD the bisector of $\angle EAC$. Let M be the midpoint of CF, and let X be chosen so that $AMXE$ is a parallelogram ($AM \parallel EX$ and $AE \parallel MX$).

Prove that BD, FX, and ME are concurrent.

Problem 2016.2. Find all positive integers n for which the cells of an $n \times n$ array can be filled with the letters I, M, O so that

(i) in every row and column, exactly one third of the entries are each of I, M, and O;

(ii) on every diagonal containing a multiple of 3 cells, exactly one third of its entries are each of I, M, and O.

Here a cell is indexed by (i, j), where $1 \leq i, j \leq n$. A diagonal is the set of cells satisfying $i + j = c$ or $i - j = c$ for some constant c; thus both types of diagonals are included.

Problem 2016.3. Let $P = A_1A_2\cdots A_k$ be a convex polygon whose vertices have integer coordinates and which is inscribed in a circle. Suppose that n is an odd positive integer and that the squares of the side lengths of P are all integers divisible by n. If S is the area of P, prove that $2S$ is an integer divisible by n.

Problem 2016.4. A set of positive integers is called *fragrant* if it has at least two elements and each element shares a prime divisor with some other element of the set. Define $P(n) = n^2+n+1$. Determine the least positive integer b for which there exists a nonnegative integer a such that

$$\{P(a+1), P(a+2), \ldots, P(a+b)\}$$

is fragrant.

Problem 2016.5. On a board is written the equation

$$(x-1)(x-2)\cdots(x-2016) = (x-1)(x-2)\cdots(x-2016),$$

with 2016 linear factors on each side. Find the smallest k such that deleting exactly k of these 4032 linear factors leaves at least one factor on each side and produces an equation with no real solution.

Problem 2016.6. There are $n \geq 2$ line segments in the plane. Every two segments cross, and no three meet at one point. Geoff places a frog at one endpoint of each segment, facing the other endpoint. He then claps $n-1$ times. At every clap, each frog jumps forward to the next intersection point along its segment, without changing direction.

Geoff wants no two frogs ever to occupy the same intersection point at the same time.

(a) Prove that Geoff can always achieve this when n is odd.

(b) Prove that Geoff can never achieve this when n is even.

Problem 2017.1. For an integer $a_0 > 1$, define the sequence $(a_n)_{n\geq 0}$ by

$$a_{n+1} = \begin{cases} \sqrt{a_n}, & \text{if } \sqrt{a_n} \text{ is an integer,} \\ a_n + 3, & \text{otherwise} \end{cases} \qquad (n \geq 0).$$

Find all a_0 for which some number A occurs as a_n for infinitely many n.

Problem 2017.2. Determine all functions $f\colon \mathbb{R} \to \mathbb{R}$ satisfying

$$f(f(x)f(y)) + f(x+y) = f(xy)$$

for all $x, y \in \mathbb{R}$.

Problem 2017.3. A hunter and an invisible rabbit play a game in the Euclidean plane. They begin at the same point, $B_0 = A_0$. After $n-1$ rounds, the rabbit and the hunter are at A_{n-1} and B_{n-1}. The following events take place in order in round n:

(1) The rabbit moves invisibly to A_n, exactly one unit from A_{n-1}.

(2) A tracking device announces a point P_n to the hunter, with the sole guarantee that $P_nA_n \leq 1$.

(3) The hunter moves visibly to B_n, exactly one unit from B_{n-1}.

Can the hunter always choose her moves so that $A_{10^9}B_{10^9} \leq 100$, regardless of the rabbit's movements and the device's reports?

Problem 2017.4. Let R, S be distinct points on a circle Ω with RS not a diameter, and let ℓ be the tangent to Ω at R. Point T satisfies S being the midpoint of RT. Choose J on the shorter arc RS of Ω such that the circumcircle Γ of JST intersects ℓ in two distinct points. Let A be the intersection of Γ and ℓ closer to R, and let AJ meet Ω again at K.

Prove that KT is tangent to Γ.

Problem 2017.5. Let $N \geq 2$ be an integer. A row contains $N(N+1)$ soccer players, all of different heights. Sir Alex removes $N(N-1)$ of them, leaving $2N$ players in their original order. Prove that the remaining players can be chosen so that the tallest two are adjacent, the third- and fourth-tallest are adjacent, and so on, with the shortest two also adjacent.

Problem 2017.6. An ordered pair $(x, y) \in \mathbb{Z}^2$ is called a *primitive point* if $\gcd(x, y) = 1$. For any finite set S of primitive points, prove that there exist a positive integer n and integers $a_0, a_1, \ldots, a_n$ such that

$$a_0 x^n + a_1 x^{n-1} y + a_2 x^{n-2} y^2 + \cdots + a_{n-1} x y^{n-1} + a_n y^n = 1$$

for every $(x, y) \in S$.

Problem 2018.1. Let Γ be the circumcircle of an acute triangle ABC. Points $D \in AB$ and $E \in AC$ satisfy $AD = AE$. The perpendicular bisectors of BD and CE meet the minor arcs AB and AC of Γ at F and G, respectively. Prove that DE and FG are parallel or coincide.

Problem 2018.2. Determine all integers $n \geq 3$ for which real numbers $a_1, a_2, \ldots, a_{n+2}$ exist with $a_{n+1} = a_1$, $a_{n+2} = a_2$, and

$$a_i a_{i+1} + 1 = a_{i+2} \qquad (i = 1, 2, \ldots, n).$$

Problem 2018.3. An *anti-Pascal triangle* is an equilateral triangular array of numbers in which every entry outside the bottom row equals the absolute difference of the two entries directly below it. For example, the following anti-Pascal triangle of four rows contains all integers from 1 to 10:

$$\begin{array}{ccccccc} & & & 4 & & & \\ & & 2 & & 6 & & \\ & 5 & & 7 & & 1 & \\ 8 & & 3 & & 10 & & 9 \end{array}$$

Does an anti-Pascal triangle with 2018 rows exist that contains every integer from 1 to $1 + 2 + \cdots + 2018$?

Problem 2018.4. A *site* is a point (x, y) with $x, y \in \{1, 2, \ldots, 20\}$. All 400 sites start unoccupied. Amy and Ben alternate placing stones, with Amy first. Amy places a red stone on an empty site, subject to the rule that no two red stones are at distance $\sqrt{5}$ from one another. Ben places a blue stone on any empty site, with no distance restriction involving blue stones. The game ends as soon as a player cannot place a stone.

Find the greatest integer K for which Amy can ensure she places at least K red stones, regardless of Ben's moves.

Problem 2018.5. Let $(a_n)_{n \geq 1}$ be a sequence of positive integers. Suppose there exists an integer $N \geq 2$ such that

$$\frac{a_1}{a_2} + \frac{a_2}{a_3} + \cdots + \frac{a_{n-1}}{a_n} + \frac{a_n}{a_1}$$

is an integer for all $n \geq N$. Prove that some positive integer M satisfies $a_m = a_{m+1}$ for every $m \geq M$.

Problem 2018.6. A convex quadrilateral $ABCD$ satisfies $AB \cdot CD = BC \cdot DA$. A point X inside $ABCD$ satisfies

$$\angle XAB = \angle XCD, \qquad \angle XBC = \angle XDA.$$

Prove that $\angle BXA + \angle DXC = 180°$.

Problem 2019.1. Find all functions $f\colon \mathbb{Z} \to \mathbb{Z}$ such that

$$f(2a) + 2f(b) = f(f(a+b))$$

for all integers a, b.

Problem 2019.2. In triangle ABC, let $A_1 \in BC$ and $B_1 \in AC$. Choose $P \in AA_1$ and $Q \in BB_1$ with $PQ \parallel AB$. Point P_1 lies on line PB_1, with B_1 strictly between P and P_1, and $\angle PP_1C = \angle BAC$. Similarly, point Q_1 lies on line QA_1, with A_1 strictly between Q and Q_1, and $\angle CQ_1Q = \angle CBA$.

Prove that P, Q, P_1, Q_1 are concyclic.

Problem 2019.3. A social network has 2019 users with mutual friendships. The following operation may be performed repeatedly: choose users A, B, C such that A is friends with both B and C, while B and C are not friends. Make B and C friends and end the friendships AB and AC, leaving every other friendship unchanged.

Initially, 1010 users each have 1009 friends, and the other 1009 users each have 1010 friends. Prove that some sequence of the allowed operations leaves every user with at most one friend.

Problem 2019.4. Find all pairs (k, n) of positive integers satisfying

$$k! = (2^n - 1)(2^n - 2)(2^n - 4) \cdots (2^n - 2^{n-1}).$$

Problem 2019.5. The Bank of Bath issues coins marked H on one side and T on the other. Harry arranges n such coins in a row and repeats this operation: when exactly $k > 0$ coins show H, he flips the k-th coin from the left; if no coins show H, he stops. For instance, starting with THT for $n = 3$ yields

$$THT \to HHT \to HTT \to TTT,$$

so the process ends after three operations.

(a) Prove that the process terminates for every initial configuration.

(b) For a configuration C, let $L(C)$ count the operations before termination. Thus $L(THT) = 3$ and $L(TTT) = 0$. Determine the average of $L(C)$ over all 2^n initial configurations C.

Problem 2019.6. Let I be the incentre of an acute triangle ABC with $AB \neq AC$. Its incircle ω touches BC, CA, and AB at D, E, and F, respectively. The line through D perpendicular to EF meets ω again at R, and AR meets ω again at P. The circumcircles of triangles PCE and PBF meet again at Q.

Prove that DI and PQ intersect on the line through A perpendicular to AI.

Problem 2020.1. Consider the convex quadrilateral $ABCD$. The point P is in the interior of $ABCD$. The following ratio equalities hold:

$$\angle PAD : \angle PBA : \angle DPA = 1 : 2 : 3 = \angle CBP : \angle BAP : \angle BPC.$$

Prove that the following three lines meet in a point: the internal bisectors of angles $\angle ADP$ and $\angle PCB$ and the perpendicular bisector of segment AB.

Problem 2020.2. The real numbers a, b, c, d are such that $a \geq b \geq c \geq d > 0$ and $a+b+c+d = 1$. Prove that

$$(a + 2b + 3c + 4d)a^a b^b c^c d^d < 1.$$

Problem 2020.3. There are $4n$ pebbles of weights $1, 2, 3, \ldots, 4n$. Each pebble is coloured in one of n colours and there are four pebbles of each colour. Show that we can arrange the pebbles into two piles so that the following two conditions are both satisfied:

- The total weights of both piles are the same.
- Each pile contains two pebbles of each colour.

Problem 2020.4. There is an integer $n > 1$. There are n^2 stations on a slope of a mountain, all at different altitudes. Each of two cable car companies, A and B, operates k cable cars; each cable car provides a transfer from one of the stations to a higher one (with no intermediate stops). The k cable cars of A have k different starting points and k different finishing points, and a cable car which starts higher also finishes higher. The same conditions hold for B. We say that two stations are *linked* by a company if one can start from the lower station and reach the higher one by using one or more cars of that company (no other movements between stations are allowed).

Determine the smallest positive integer k for which one can guarantee that there are two stations that are linked by both companies.

Problem 2020.5. A deck of $n > 1$ cards is given. A positive integer is written on each card. The deck has the property that the arithmetic mean of the numbers on each pair of cards is also the geometric mean of the numbers on some collection of one or more cards.

For which n does it follow that the numbers on the cards are all equal?

Problem 2020.6. Prove that there exists a positive constant c such that the following statement is true:

Consider an integer $n > 1$, and a set S of n points in the plane such that the distance between any two different points in S is at least 1. It follows that there is a line ℓ separating S such that the distance from any point of S to ℓ is at least $cn^{-1/3}$.

(A line ℓ *separates* a set of points S if some segment joining two points in S crosses ℓ.)

Problem 2021.1. Let $n \geq 100$ be an integer. Ivan writes the numbers $n, n+1, \ldots, 2n$ each on different cards. He then shuffles these $n+1$ cards, and divides them into two piles. Prove that at least one of the piles contains two cards such that the sum of their numbers is a perfect square.

Problem 2021.2. Show that the inequality

$$\sum_{i=1}^{n}\sum_{j=1}^{n}\sqrt{|x_i - x_j|} \leq \sum_{i=1}^{n}\sum_{j=1}^{n}\sqrt{|x_i + x_j|}$$

holds for all real numbers $x_1, \ldots, x_n$.

Problem 2021.3. Let D be an interior point of the acute triangle ABC with $AB > AC$ so that $\angle DAB = \angle CAD$. The point E on the segment AC satisfies $\angle ADE = \angle BCD$, the point F on the segment AB satisfies $\angle FDA = \angle DBC$, and the point X on the line AC satisfies $CX = BX$. Let O_1 and O_2 be the circumcentres of the triangles ADC and EXD, respectively. Prove that the lines BC, EF, and O_1O_2 are concurrent.

Problem 2021.4. Let Γ be a circle with centre I, and $ABCD$ a convex quadrilateral such that each of the segments AB, BC, CD and DA is tangent to Γ. Let Ω be the circumcircle of the triangle AIC. The extension of BA beyond A meets Ω at X, and the extension of BC beyond C meets Ω at Z. The extensions of AD and CD beyond D meet Ω at Y and T, respectively. Prove that

$$AD + DT + TX + XA = CD + DY + YZ + ZC.$$

Problem 2021.5. Two squirrels, Bushy and Jumpy, have collected 2021 walnuts for the winter. Jumpy numbers the walnuts from 1 through 2021, and digs 2021 little holes in a circular pattern in the ground around their favourite tree. The next morning Jumpy notices that Bushy had placed one walnut into each hole, but had paid no attention to the numbering. Unhappy, Jumpy decides to reorder the walnuts by performing a sequence of 2021 moves. In the k-th move, Jumpy swaps the positions of the two walnuts adjacent to walnut k.

Prove that there exists a value of k such that, on the k-th move, Jumpy swaps some walnuts a and b such that $a < k < b$.

Problem 2021.6. Let $m \geq 2$ be an integer, A be a finite set of (not necessarily positive) integers, and $B_1, B_2, B_3, \ldots, B_m$ be subsets of A. Assume that for each $k = 1, 2, \ldots, m$ the sum of the elements of B_k is m^k. Prove that A contains at least $m/2$ elements.

Problem 2022.1. The Bank of Oslo issues two types of coin: aluminium (denoted A) and bronze (denoted B). Marianne has n aluminium coins and n bronze coins, arranged in a row in some arbitrary initial order. A *chain* is any subsequence of consecutive coins of the same type. Given a fixed positive integer $k \leq 2n$, Marianne repeatedly performs the following operation: she identifies the longest chain containing the k-th coin from the left, and moves all coins in that chain to the left end of the row. For example, if $n = 4$ and $k = 4$, the process starting from the ordering $AABBBABA$ would be

$$AAB\underline{B}BABA \to BBB\underline{A}AABA \to AAA\underline{B}BBBA$$
$$\to BBB\underline{B}AAAA \to BBB\underline{B}AAAA \to \cdots.$$

Find all pairs (n, k) with $1 \leq k \leq 2n$ such that for every initial ordering, at some moment during the process, the leftmost n coins will all be of the same type.

Problem 2022.2. Find all functions $f\colon \mathbb{R}_{>0} \to \mathbb{R}_{>0}$ such that for each $x \in \mathbb{R}_{>0}$, there is exactly one $y \in \mathbb{R}_{>0}$ satisfying

$$xf(y) + yf(x) \leq 2.$$

Problem 2022.3. Let k be a positive integer and let S be a finite set of odd prime numbers. Prove that there is at most one way (up to rotation and reflection) to place the elements of S around a circle such that the product of any two neighbours is of the form $x^2 + x + k$ for some positive integer x.

Problem 2022.4. Let $ABCDE$ be a convex pentagon such that $BC = DE$. Assume that there is a point T inside $ABCDE$ with $TB = TD$, $TC = TE$ and $\angle ABT = \angle TEA$. Let line AB intersect lines CD and CT at points P and Q, respectively. Assume that the points P, B, A, Q occur on their line in that order. Let line AE intersect lines CD and DT at points R and S, respectively. Assume that the points R, E, A, S occur on their line in that order. Prove that the points P, S, Q, R lie on a circle.

Problem 2022.5. Find all triples (a, b, p) of positive integers with p prime and

$$a^p = b! + p.$$

Problem 2022.6. Let n be a positive integer. A *Nordic square* is an $n \times n$ board containing all the integers from 1 to n^2 so that each cell contains exactly one number. Two different cells are considered adjacent if they share a common side. Every cell that is adjacent only to cells containing larger numbers is called a *valley*. An *uphill path* is a sequence of one or more cells such that:

(i) the first cell in the sequence is a valley,
(ii) each subsequent cell in the sequence is adjacent to the previous cell, and
(iii) the numbers written in the cells in the sequence are in increasing order.

Find, as a function of n, the smallest possible total number of uphill paths in a Nordic square.

Problem 2023.1. Determine all composite integers $n > 1$ that satisfy the following property: if $d_1, d_2, \ldots, d_k$ are all the positive divisors of n with $1 = d_1 < d_2 < \cdots < d_k = n$, then d_i divides $d_{i+1} + d_{i+2}$ for every $1 \leq i \leq k - 2$.

Problem 2023.2. Let ABC be an acute-angled triangle with $AB < AC$. Let Ω be the circumcircle of ABC. Let S be the midpoint of the arc CB of Ω containing A. The perpendicular from A to BC meets BS at D and meets Ω again at $E \neq A$. The line through D parallel to BC meets line BE at L. Denote the circumcircle of triangle BDL by ω. Let ω meet Ω again at $P \neq B$.

Prove that the line tangent to ω at P meets line BS on the internal angle bisector of $\angle BAC$.

Problem 2023.3. For each integer $k \geq 2$, determine all infinite sequences of positive integers $a_1, a_2, \ldots$ for which there exists a polynomial P of the form $P(x) = x^k + c_{k-1}x^{k-1} + \cdots + c_1 x + c_0$, where $c_0, c_1, \ldots, c_{k-1}$ are non-negative integers, such that

$$P(a_n) = a_{n+1}a_{n+2} \cdots a_{n+k}$$

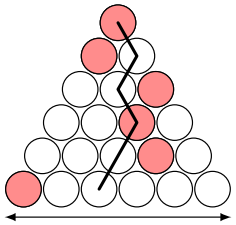
for every integer $n \geq 1$.

Problem 2023.4. Let $x_1, x_2, \ldots, x_{2023}$ be pairwise different positive real numbers such that

$$a_n = \sqrt{(x_1 + x_2 + \cdots + x_n)\left(\frac{1}{x_1} + \frac{1}{x_2} + \cdots + \frac{1}{x_n}\right)}$$

is an integer for every $n = 1, 2, \ldots, 2023$. Prove that $a_{2023} \geq 3034$.

Problem 2023.5. Let n be a positive integer. A *Japanese triangle* consists of $1 + 2 + \cdots + n$ circles arranged in an equilateral triangular shape such that for each $i = 1, 2, \ldots, n$, the i-th row contains exactly i circles, exactly one of which is coloured red. A *ninja path* in a Japanese triangle is a sequence of n circles obtained by starting in the top row, then repeatedly going from a circle to one of the two circles immediately below it and finishing in the bottom row. Here is an example of a Japanese triangle with $n = 6$, along with a ninja path in that triangle containing two red circles.



In terms of n, find the greatest k such that in each Japanese triangle there is a ninja path containing at least k red circles.

Problem 2023.6. Let ABC be an equilateral triangle. Let A_1, B_1, C_1 be interior points of ABC such that $BA_1 = A_1C$, $CB_1 = B_1A$, $AC_1 = C_1B$, and

$$\angle BA_1C + \angle CB_1A + \angle AC_1B = 480^\circ.$$

Let BC_1 and CB_1 meet at A_2, let CA_1 and AC_1 meet at B_2, and let AB_1 and BA_1 meet at C_2. Prove that if triangle $A_1B_1C_1$ is scalene, then the three circumcircles of triangles AA_1A_2, BB_1B_2 and CC_1C_2 all pass through two common points.

Problem 2024.1. Determine all real numbers α such that, for every positive integer n, the integer

$$\lfloor \alpha \rfloor + \lfloor 2\alpha \rfloor + \cdots + \lfloor n\alpha \rfloor$$

is a multiple of n.

Problem 2024.2. Determine all pairs (a, b) of positive integers for which there exist positive integers g and N such that

$$\gcd(a^n + b, b^n + a) = g$$

holds for all integers $n \geq N$.

Problem 2024.3. Let $a_1, a_2, a_3, \ldots$ be an infinite sequence of positive integers, and let N be a positive integer. Suppose that, for each $n > N$, a_n is equal to the number of times a_{n-1} appears in the list $a_1, a_2, \ldots, a_{n-1}$.

Prove that at least one of the sequences $a_1, a_3, a_5, \ldots$ and $a_2, a_4, a_6, \ldots$ is eventually periodic.

(An infinite sequence $b_1, b_2, b_3, \ldots$ is *eventually periodic* if there exist positive integers p and M such that $b_{m+p} = b_m$ for all $m \geq M$.)

Problem 2024.4. Let ABC be a triangle with $AB < AC < BC$. Let the incentre and incircle of triangle ABC be I and ω, respectively. Let X be the point on line BC different from C such that the line through X parallel to AC is tangent to ω. Similarly, let Y be the point on line BC different from B such that the line through Y parallel to AB is tangent to ω. Let AI intersect the circumcircle of triangle ABC again at $P \neq A$. Let K and L be the midpoints of AC and AB, respectively.

Prove that $\angle KIL + \angle YPX = 180^\circ$.

Problem 2024.5. Turbo the snail plays a game on a board with 2024 rows and 2023 columns. There are hidden monsters in 2022 of the cells. Initially, Turbo does not know where any of the monsters are, but he knows that there is exactly one monster in each row except the first row and the last row, and that each column contains at most one monster.

Turbo makes a series of attempts to go from the first row to the last row. On each attempt, he chooses to start on any cell in the first row, then repeatedly moves to an adjacent cell sharing a common side. (He is allowed to return to a previously visited cell.) If he reaches a cell with a

monster, his attempt ends and he is transported back to the first row to start a new attempt. The monsters do not move, and Turbo remembers whether or not each cell he has visited contains a monster. If he reaches any cell in the last row, his attempt ends and the game is over.

Determine the minimum value of n for which Turbo has a strategy that guarantees reaching the last row on the n-th attempt or earlier, regardless of the locations of the monsters.

Problem 2024.6. A function $f\colon \mathbb{Q} \to \mathbb{Q}$ is called *aquaesulian* if the following property holds: for every $x, y \in \mathbb{Q}$,

$$f(x + f(y)) = f(x) + y \qquad \text{or} \qquad f(f(x) + y) = x + f(y).$$

Show that there exists an integer c such that for any aquaesulian function f there are at most c different rational numbers of the form $f(r) + f(-r)$ for some rational number r, and find the smallest possible value of c.

Problem 2025.1. A line in the plane is called *sunny* if it is not parallel to any of the x-axis, the y-axis, and the line $x + y = 0$.

Let $n \geq 3$ be a given integer. Determine all nonnegative integers k such that there exist n distinct lines in the plane satisfying both of the following:

- for all positive integers a and b with $a + b \leq n + 1$, the point (a, b) is on at least one of the lines; and
- exactly k of the n lines are sunny.

Problem 2025.2. Let Ω and Γ be circles with centres M and N, respectively, such that the radius of Ω is less than the radius of Γ. Suppose circles Ω and Γ intersect at two distinct points A and B. Line MN intersects Ω at C and Γ at D, such that points C, M, N and D lie on the line in that order. Let P be the circumcentre of triangle ACD. Line AP intersects Ω again at $E \neq A$. Line AP intersects Γ again at $F \neq A$. Let H be the orthocentre of triangle PMN.

Prove that the line through H parallel to AP is tangent to the circumcircle of triangle BEF.

Problem 2025.3. A function $f\colon \mathbb{N} \to \mathbb{N}$ is said to be *bonza* if

$$f(a) \mid b^a - f(b)^{f(a)}$$

for all positive integers a and b.

Determine the smallest real constant c such that $f(n) \leq cn$ for all bonza functions f and all positive integers n.

Problem 2025.4. A *proper divisor* of a positive integer N is a positive divisor of N other than N itself.

The infinite sequence $a_1, a_2, \ldots$ consists of positive integers, each of which has at least three proper divisors. For each $n \geq 1$, the integer a_{n+1} is the sum of the three largest proper divisors of a_n.

Determine all possible values of a_1.

Problem 2025.5. Alice and Bazza are playing the *inekoalaty game*, a two-player game whose rules depend on a positive real number λ which is known to both players. On the n-th turn of the game (starting with $n = 1$) the following happens:

- If n is odd, Alice chooses a nonnegative real number x_n such that

$$x_1 + x_2 + \cdots + x_n \leq \lambda n.$$

- If n is even, Bazza chooses a nonnegative real number x_n such that

$$x_1^2 + x_2^2 + \cdots + x_n^2 \leq n.$$

If a player cannot choose a suitable number x_n, the game ends and the other player wins. If the game goes on forever, neither player wins. All chosen numbers are known to both players.

Determine all values of λ for which Alice has a winning strategy and all those for which Bazza has a winning strategy.

Problem 2025.6. Consider a 2025×2025 grid of unit squares. Matilda wishes to place on the grid some rectangular tiles, possibly of different sizes, such that each side of every tile lies on a grid line and every unit square is covered by at most one tile.

Determine the minimum number of tiles Matilda needs to place so that each row and each column of the grid has exactly one unit square that is not covered by any tile.

Problem 2026.1. There are 2026 integers greater than 1 written on a blackboard, not necessarily different. In a move, Confucius chooses two integers $m > 1$ and $n > 1$ from different places on the blackboard and replaces these two integers with

$$\gcd(m, n) \qquad \text{and} \qquad \frac{\operatorname{lcm}(m, n)}{\gcd(m, n)}.$$

He continues to make moves while it is possible to do so.

(a) Prove that, regardless of the choices of Confucius, after finitely many moves, exactly one integer M on the blackboard is greater than 1.

(b) Prove that the value of M does not depend on the choices of Confucius.

Problem 2026.2. Let ABC be a triangle and let points M and N be the midpoints of sides AB and AC, respectively. Let points K and L be chosen strictly inside triangles BMC and BNC, respectively, such that K lies strictly inside triangle ABL and L lies strictly inside triangle AKC. Suppose that

$$\angle KBA = \angle ACL, \qquad \angle LBK = \angle LNC, \qquad \angle LCK = \angle BMK.$$

Let O be the circumcentre of triangle AKL. Prove that $OM = ON$.

Problem 2026.3. Let n be a positive integer. Liu Bang and Xiang Yu have a stick of length 1 and want to divide it between themselves. Liu marks at most n points on the stick, and then Xiang marks at most n points on the stick. The marked points are distinct. Then, the stick is cut at all marked points, creating a number of pieces. Afterwards, they take turns claiming any unclaimed piece of the stick, with Liu going first. Each player's goal is to maximise the total length of their own pieces.

For each n, determine the largest value c such that Liu may guarantee a total length of at least c, regardless of Xiang's play.

Problem 2026.4. Shan-Yu and Mulan are playing a game. Let θ be an angle with $0^\circ < \theta < 180^\circ$ known to both players. Initially, Shan-Yu makes a paper triangle $\mathcal{T}$ with measurements of his choice. Then, they repeatedly perform the following steps:

- If $\mathcal{T}$ has at least one angle measuring exactly θ, then the game stops and Mulan wins.
- Otherwise, Mulan chooses a point P on the perimeter of $\mathcal{T}$, different from its three vertices. She then makes a straight cut from P to the opposite vertex of $\mathcal{T}$, splitting it into two triangles.
- Shan-Yu discards one of the two triangles. The remaining triangle becomes the new $\mathcal{T}$.

For which real values of θ can Mulan guarantee her victory in finitely many steps, no matter how Shan-Yu plays?

Problem 2026.5. Determine all functions $f\colon \mathbb{R}_{>0} \to \mathbb{R}_{>0}$ such that

$$\sqrt{\frac{x^2 + f(y)^2}{2}} \geq \frac{f(x) + y}{2} \geq \sqrt{xf(y)}$$

for every $x, y \in \mathbb{R}_{>0}$.

Problem 2026.6. Let $a_1, a_2, a_3, \ldots$ be an infinite sequence of positive integers greater than 1. Suppose that for all positive integers n, the number a_{n+1} is the smallest positive integer greater than a_n such that $\gcd(a_{n+1}, a_i) > 1$ for every $i = 1, 2, \ldots, n$. Prove that there exist positive integers T and L such that

$$a_{n+T} = a_n + L$$

for every positive integer n.

Part II. Shortlisted Problems